

Chapter 64 — Paediatric Urological Oncology: Pelvis, Bladder, Testis and Genitalia

- Tumours of the pelvis, bladder, gonads and genitalia are **rare in children**. **Rhabdomyosarcoma is the commonest**, but the differential is long and spans benign, malignant and intermediate-biology tumours.
- **Children with cancer do not present only to high-volume cancer centres**, so every paediatric urologist needs working knowledge of these tumours and their management.

RHABDOMYOSARCOMA

- Rhabdomyosarcoma (RMS) is an **aggressive soft-tissue sarcoma derived from rhabdomyoblasts** (striated-muscle precursor cells). Because striated muscle is ubiquitous, it can arise almost anywhere; the commonest sites are the head and neck, the extremities and the genitourinary (GU) tract.
- **Five-year survival for GU-RMS exceeds 80%** — below Wilms tumour (90% or better) but above RMS at non-GU sites (about 70%).
- The GU sites are **bladder/prostate (BP), paratesticular (PT) and vulvovaginal (VV)**.

A Brief History of Rhabdomyosarcoma Collaborative Research

- The **Intergroup Rhabdomyosarcoma Study Group (IRSG)** formed in 1972; survival improved as a direct result of collaboration. **Survival was below 25% before consistent multiagent chemotherapy**, popularised by IRS-I (Intergroup Rhabdomyosarcoma Study I).
- **Overall survival (OS) for localised RMS rose from 55% (IRS-I) to 71% (IRS-IV)** through better staging, risk stratification, improved local therapy and better supportive care.
- Protocols (more than 5200 children): **IRS-I 1972–1978 → IRS-II 1978–1984 → IRS-III 1984–1991 → IRS-IV Pilot (advanced disease only) 1987–1991 → IRS-IV 1991–1997 → IRS-V, the “D-series” (D9803 intermediate risk, D9602 low risk), 1997–2005**.
- **In 2000, IRSG merged** with the Pediatric Oncology Group, the National Wilms Tumor Study Group and the Children’s Cancer Group to form the **Children’s Oncology Group (COG)**; IRSG became the COG Soft Tissue Sarcoma Committee, which took over D9803 and D9602. COG is affiliated with more than 200 hospitals; **over 90% of North American paediatric cancer patients are treated at a COG facility**.
- Europe has several groups, including single-nation groups in the UK, France and Italy.

RHABDOMYOSARCOMA COOPERATIVE GROUPS COMPARED

Group	Formed	Constitution	Distinguishing feature
IRSG → COG	1972 → 2000	North American; IRSG merged into COG	Ran the IRS-I to IRS-V trials
EpSSG (European paediatric Soft tissue sarcoma Study Group)	2005	Founded by SIOP-MMT (International Society of Paediatric Oncology — Malignant Mesenchymal Tumour Committee) and AIEOP (Associazione Italiana Ematologia Oncologia Pediatrica) Soft Tissue Sarcoma Committee	Pan-European successor to national groups
CWS (Cooperative Weichteilsarkom Studiengruppe)	1980	Predominantly German-speaking: Germany, Austria, Poland, Sweden, Switzerland, Hungary	Several relatively large RMS trials

Group	Formed	Constitution	Distinguishing feature
INSTRuCT (INternational Soft Tissue SaRcoma ConsorTium)	Most recent	Data and collaboration from COG, EpSSG and CWS, plus historical IRSG, SIOP-MMT and AIEOP data	Common dataset to harmonise staging, risk factors and consensus treatment

Concept — RMS outcomes were won by **protocol-driven multicentre collaboration**, not by any single operation; the current frontier is harmonising the competing international systems into one dataset.

Epidemiology

- **Commonest sarcoma of childhood and adolescence**, about 3% of all paediatric cancers.
- In the United States **400–500 children develop RMS each year**; **20–30% arise in GU organs**.
- Age-adjusted incidence about 4.6 per million, **stable for decades**.
- Most cases occur in the first decade, with a **bimodal distribution: a major peak in the first 2 years and a smaller adolescent peak**.
- Survival is slightly higher in non-Hispanic White than non-Hispanic Black patients; a socioeconomic survival gradient was **not statistically significant**.

Histopathology and Molecular Characterisation

- Four histological subtypes exist; **only embryonal (ERMS) and alveolar (ARMS) typically occur in GU organs**. Pleomorphic and spindle cell/sclerosing RMS are rare in children, particularly at GU sites.
- Local infiltration and regional nodal spread are common; **distant metastases are present in 16% at diagnosis** (lung, bone, bone marrow).
- The fusion is a translocation between **PAX3 (chromosome 2) or PAX7 (chromosome 1) and FOXO1 (chromosome 13)**, producing PAX3/7-FOXO1.

EMBRYONAL VERSUS ALVEOLAR RHABDOMYOSARCOMA

Feature	Embryonal (ERMS)	Alveolar (ARMS)
Usual site	GU tract	Trunk, extremities
Age	Younger; about 70% under 10 years	More even age distribution; older children, adolescents
Fusion status	Typically fusion-negative	About 80% PAX3/7-FOXO1 fusion-positive
Syndromic RMS	Predominant subtype	—
Prognosis	Better	Fusion-positive: worse event-free survival (EFS) and OS; fusion-negative ARMS behaves like ERMS

- Because fusion status predicts outcome better than histology, **current collaborative trials assign risk group by fusion status rather than histology**.

KEY POINTS

- About a quarter of new RMS diagnoses arise in GU organs; OS for GU-RMS exceeds 80%.
- COG, EpSSG and CWS have dramatically improved care and survival.
- The commonest GU-RMS histology is embryonal.

Concept — **the fusion gene, not the microscope, defines the biology**: fusion-negative alveolar tumours behave as embryonal ones, which is why histology has been displaced from risk assignment.

Syndromic Associations

- **Most RMS is sporadic; 5% or more is linked to a germline syndrome, and syndromic RMS is more often ERMS.**

CANCER PREDISPOSITION SYNDROMES ASSOCIATED WITH RHABDOMYOSARCOMA

Syndrome	Genetic basis	Tumour associations	Surveillance
Li-Fraumeni	Germline TP53 (p53 tumour suppressor)	Soft-tissue sarcoma, osteosarcoma, early-onset breast cancer, brain tumours, adrenocortical carcinoma; cancer risk 50% by age 30, up to 100% in females by 70; RMS (especially ARMS) in the first decade	Abdominal ultrasound (US) every 3–4 months; annual whole-body and brain magnetic resonance imaging (MRI) after age 1
Beckwith-Wiedemann (BWS)	Aberrant DNA methylation at 11p15.5 imprinting region	Classically Wilms tumour; RMS in up to 5%	Abdominal US and alpha-fetoprotein (AFP) every 3 months to age 4; renal US every 3 months age 4–7; benefit beyond 8 unclear
DICER1	Heterozygous germline loss-of-function in the microRNA-processing gene DICER1; familial pleuropulmonary blastoma (PPB)	RMS, cystic nephroma, Wilms tumour, ovarian Sertoli-Leydig cell tumours	Abdominal US and chest computed tomography (CT, for PPB) in infancy, then every 6–12 months to at least age 8; annual US to 12; pelvic US to 40 (some authors) for ovarian stromal tumours
Neurofibromatosis type 1 (NF-1)	Autosomal dominant RASopathy (Ras deregulation)	20 times over-represented among IRS-IV enrollees; lifetime RMS risk 5–10%, notably pelvic	RMS-specific surveillance controversial; standard NF-1 surveillance likely sufficient
Costello	Autosomal dominant RASopathy (as Noonan)	Lifetime tumour risk 13%; bladder/urothelial tumours, neuroblastoma, RMS	Abdominal and pelvic US every 3 months to age 8–10

- What the table tests: the **surveillance modality follows the dominant tumour** (whole-body MRI for the multi-organ Li-Fraumeni spectrum, chest CT for PPB in DICER1), and the **RASopathies (NF-1, Costello) carry a pelvic/bladder bias**.

Cancer Genetics

- The **most significant recent development is the discovery of oncogenic fusion proteins** from chromosomal translocations; PAX3/7-FOXO1 is the commonest and most clinically useful.
- Other fusions, of **uncertain clinical import** (INSTRuCT biobanking should clarify):

OTHER RHABDOMYOSARCOMA FUSION GENES

Class	Fusions	Subtype
PAX3/7 partners	FOXO4, INO80D, NCOA1, NCOA2	ARMS and/or ERMS; variable prognostic impact
Non-PAX	CAV1-MET, EP300-VGLL3, HMGA2-NEGR1	Spindle cell RMS
Non-PAX	SRF-NCOA1, SRF-FOXO1	Well-differentiated RMS

- **Diagnosis before age 3 years is associated with a germline predisposition variant.** Commonest pathogenic germline variants in early-onset RMS: **TP53, NF1, BRCA2**; DICER1, HRAS and loss of imprinting (LOI) are also implicated — the same genes that underlie the syndromes above.
- **MYOD1** expression in about 3% of fusion-negative tumours → **significantly worse prognosis**; mostly head and neck, occasionally elsewhere.

Biomarkers

- **PAX3/7-FOXO1 fusion status is currently the most clinically meaningful biomarker.**
- **Circulating tumour DNA (ctDNA)** at diagnosis in fusion-negative intermediate-risk RMS (many BP) was associated with significantly worse EFS and OS in a COG analysis, **independent on multivariable analysis**. If prospectively confirmed, ctDNA may enter risk stratification.

Concept — risk assignment is moving **from anatomy and histology to molecular biology**: fusion status now, ctDNA and MYOD1 next; and young age at diagnosis should prompt a search for germline predisposition.

KEY POINTS

- Most RMS is sporadic; some arises from predisposition syndromes such as Beckwith-Wiedemann or DICER1.
- PAX3/7-FOXO1 fusion status is more clinically relevant than histology.

Presentation and Diagnosis

- Presentation is **determined by site**:
 - **BP-RMS**: haematuria, stranguria or frank urinary retention; botryoid tumour may protrude from the urethral meatus. **A 3- to 5-year-old boy with haematuria and new-onset stranguria needs immediate urological and imaging assessment.**
 - **PT-RMS**: palpable scrotal mass; may be found incidentally during evaluation for trauma or torsion.
 - **VV-RMS**: bloody vaginal discharge or a prolapsing vaginal/introital mass.

Initial Evaluation

- Sequence: **US first** (new mass, haematuria, obstructive symptoms) → mass suspected → **definitive CT or MRI**, including fine-cut assessment of the site's nodal basin:
 - BP: pelvic and retroperitoneal nodes **up to the renal vessels**.
 - PT: retroperitoneal nodes.
 - VV: inguinal, pelvic and retroperitoneal nodes.
- **Chest CT is mandatory** — lung is the commonest metastatic site.
- Bone scan, bone marrow biopsy and aspirates are indicated for most RMS, but **COG now recommends reduced staging in fusion-negative disease**: non-invasive (T1), small (≤ 5 cm), NO ERMS has virtually no chance of bone metastasis, so **ERMS without nodal or lung metastases needs no bone scan or marrow evaluation**.
- Positron emission tomography-CT (PET-CT): role in staging and response **controversial**; whether it could supplement or replace lymph node dissection is a question of particular interest to urologists.
- Baseline laboratory tests: full blood count, electrolytes and renal function, liver function, urinalysis.

Stage, Group and Risk Stratification

- Three layers: **pretreatment TNM (tumour, node, metastasis) stage** → **post-biopsy/resection surgical-pathological clinical group** → **composite risk group** (low, intermediate, high) from stage, group, age and fusion status.
- Stage sorts **favourable versus unfavourable sites**: in the GU tract, **PT and VV are favourable; BP is unfavourable**. Most PT and VV tumours are stage 1; **BP cannot be less than stage 2**. European collaboratives (EpSSG, CWS) use a similar system but **factor site into risk stratification rather than stage**.

- Clinical group is driven by the **completeness of surgical resection** (post-biopsy residual disease). BP-RMS was historically treated by up-front exenteration; the shift to organ preservation means **initial biopsy rather than resection, moving most BP-RMS from group I or II (R0 or R1 resection) to group III**.
- Result: **BP-RMS is most commonly intermediate risk; PT-RMS is almost always low risk**.
- **Age is an independent prognostic factor**: under 1 year and over 10 years → significantly inferior survival → higher risk group.
- International harmonisation (COG, INSTRUCT) is increasingly adding biological factors (fusion proteins) to clinical ones (age, size).

TABLE 64.1 UPDATED IRSG/COG CLINICAL GROUP CLASSIFICATION SCHEMA

Group	Extent	Resection	Residual disease / nodes
I	Localised	Complete	No regional node involvement
IIa	Localised	Gross resection	Microscopic residual; nodes not involved
IIb	Regional, involved nodes	Complete	No microscopic residual; most distal node histologically negative
IIc	Regional, involved nodes	Gross resection	Microscopic residual and/or most distal regional node involved
III	Localised or regional	Incomplete, or biopsy only	Gross residual
IV	Distant metastases at onset	—	—

- Group is **finalised at the start of chemotherapy**: a pretreatment re-excision can change group; a delayed primary excision cannot.
- Trap: a completely resected nodal specimen whose **most distal node is positive is IIc, not IIb**.

TABLE 64.2A UPDATED COG TNM PRETREATMENT STAGING SYSTEM

Stage	Sites	Size	N	M
1	Favourable: orbit, head and neck, paratesticular, female GU tract	Any	Any	M0
2	Unfavourable: parameningeal, bladder/prostate , extremity, other (trunk, retroperitoneum)	≤5 cm	N0 or NX	M0
3	Unfavourable sites (as stage 2)	≤5 cm >5 cm	N1 Any	M0
4	Any	Any	Any	M1

- Table 64.2B TNM detail: **T1** confined to site of origin; **T2** extends into or fixed to surrounding tissue. **N0** nodes not clinically involved; **N1 clinically involved** = ≥1 cm short axis on CT or MRI; **NX** unknown (typically sites that preclude nodal evaluation, e.g. cystoscopic biopsy of BP-RMS). **M0/M1** distant metastases absent/present.
- Pattern: at a favourable site **size and nodes do not raise stage**; at an unfavourable site the **5-cm and N1 thresholds** separate stage 2 from 3.

TABLE 64.3 COG RISK GROUP CATEGORISATION SYSTEM

Risk group	Stage	Clinical group	Age	PAX3/7–FOXO1
Very low	1	I	Any	Negative
Low	1	I, II; III (orbit only)	Any	Negative
Low	2	I, II	Any	Negative
Intermediate	1, 2	III (non-orbit)	Any	Negative
Intermediate	3	I, II	Any	Negative

Risk group	Stage	Clinical group	Age	PAX3/7-FOXO1
Intermediate	1, 2, 3	I, II, III	Any	Positive
Intermediate	4	IV	<10 years	Negative
High	4	IV	≥10 years	Negative
High	4	IV	Any	Positive

- Pattern: **fusion-positive disease is never low risk; metastatic disease is high risk if fusion-positive or ≥10 years**, otherwise intermediate. Age enters the table only for stage 4.
- Trap: biopsy-only (group III) BP-RMS is intermediate risk even when small and node-negative, because the **group III orbit exception does not extend to GU sites**.

KEY POINTS

- BP-RMS: haematuria or stranguria. PT-RMS: palpable scrotal mass. VV-RMS: bloody discharge or prolapsing mass.
- Tumour location, fusion status and extent of resection determine risk category, which determines therapy.

Concept — *stage records where the tumour began, group records what the surgeon left behind, and risk adds biology; organ-preserving biopsy deliberately places BP-RMS in group III and therefore intermediate risk.*

Treatment: General Principles of Rhabdomyosarcoma Treatment

Surgical principles and local control.

- **Local control = radiotherapy and/or surgery; chemotherapy is universal**, treating widespread disease and preventing recurrence.
- **Complete resection of GU-RMS (except PT-RMS) is attempted only if organ preservation is possible**; local failure and progression are the exceptions. Initial surgery is therefore usually **biopsy only**, so most BP-RMS starts chemotherapy with gross disease (group III, intermediate risk).
- The clinical group, set by the extent of resection, **determines whether radiotherapy is given and at what dose**.

PRETREATMENT RE-EXCISION VERSUS DELAYED PRIMARY EXCISION

	Pretreatment re-excision (PRE)	Delayed primary excision (DPE)
Timing	Wide local excision before chemotherapy	Wide local excision after chemotherapy has started
Survival in GU-RMS	Improves survival	Not shown to improve survival (does at other sites)
Indication	Resectable without organ compromise; microscopically positive or uncertain margins	Only if confident of R0 (negative-margin) resection, ideally without organ compromise
Clinical group	Can change group	Cannot change group
Other benefit	—	May allow radiation dose reduction

- **End-of-therapy masses** are common, particularly in BP-RMS. If **stable and asymptomatic → no cystoscopy, biopsy or excision**: biopsy often shows benign mature rhabdomyoblasts that are misread as residual tumour, and resection does not improve survival. Resect only for pain or obstruction, ideally bladder-sparing.

KEY POINTS

- PRE is excision before chemotherapy; DPE is excision after it. PRE improves survival in GU-RMS; DPE does not.
- Excising a stable end-of-therapy mass does not improve survival.

Concept — in GU-RMS **chemotherapy is the backbone and surgery serves organ preservation**; the only excision that changes outcome is the one done before chemotherapy.

Bladder/Prostate Rhabdomyosarcoma

- Biopsy sequence: **cystoscopic (preferred) → percutaneous → open transvesical** (last resort: significant risk of tumour spill).
- Cystoscopic biopsy is harder than in adult urothelial cancer: **paediatric resectoscopes are 9 or 11.5 French** with even smaller working elements. Technique: **undermine a section of tumour with cutting current → remove it intact with biopsy forceps; avoid excess cautery**, whose artefact delays and confounds the diagnosis.
- **Relieve obstruction early and effectively to protect renal function before systemic therapy.** Sequence:
 - **Urethral catheter** — effective for most; response is often rapid and the catheter can be removed within weeks.
 - Ureteric obstruction → **indwelling JJ stents**; if a large pelvic mass prevents retrograde placement → **percutaneous nephrostomy** (best option).
 - Suprapubic tube: possible, but with a primary bladder tumour **risks seeding/spillage**, as does open biopsy.
 - **Vesicostomy or ureterostomy: rarely helpful** — surgery delays chemotherapy, the treatment most likely to relieve the obstruction.
- **Residual mass after chemotherapy and local control is usually not viable tumour:** tumour cells involute leaving stroma, or differentiate into mature rhabdomyoblasts. Observe; resection does not improve survival; **second-look or extirpative surgery should be discouraged.**
- By contrast, **high-risk patients with viable tumour and no distant metastasis after chemoradiation** may be cured by extirpative or exenterative surgery. **Partial cystectomy** suits local/regional disease after chemoradiation or early treatment failure if an adequate margin is possible without unduly compromising capacity and function.
- Primary radical cystoprostatectomy/exenteration declined from **virtually all patients in IRS-I to 13% in IRS-IV, with OS unchanged (78–83%).**
- **Only 40% of IRS-IV patients who kept their bladder had “normal” bladder function;** objective assessment (validated questionnaires, urodynamics) is scarce, so the true incidence is unknown. **Detrusor overactivity and reduced functional bladder capacity** are common; anticholinergics help frequency, urgency and incontinence (anecdotal).
- **Follow-up: baseline urodynamic study (UDS) 6 months after therapy → annual renal US for at least 10 years;** repeat UDS and imaging if urinary symptoms arise.

Paratesticular Rhabdomyosarcoma

- Typically a mass within the tunica vaginalis but separate from the testis. **Treat by radical orchidectomy via an inguinal approach.**

- If a **scrotal approach** was mistakenly used → **excise the remaining cord via an inguinal incision as a PRE before chemotherapy**. Hemiscrotectomy or scrotal radiation has not improved outcome and should be avoided — consider them only for a **positive cord margin or direct scrotal skin invasion**.
- **Tumour abutting but not invading the tunica vaginalis is not a positive margin**; no further local treatment is needed.
- **Nerve-sparing staging ipsilateral retroperitoneal lymph node dissection (RPLND) before chemotherapy** if age ≥ 10 years, or at **any age with retroperitoneal lymphadenopathy (N1) on imaging**. Sample at least **7–13 nodes**; the **upper limit is the renal vessels**.
- Rationale (COG, SIOP-MMT, CWS, EpSSG data and international consensus): patients ≥ 10 years are **much more likely to have retroperitoneal disease**, and **CT has a very high false-negative rate** for retroperitoneal nodes.
- No modality is proven superior; the authors prefer **robotic RPLND** (3–4 robotic ports and a 12-mm AirSeal assistant port) — anecdotally shorter stay, faster recovery and fewer chemotherapy delays.

Concept — *PT-RMS is the one GU site where resection is always done, because organ preservation is not the constraint there; the only question is nodal staging, and age ≥ 10 replaces imaging because CT misses nodal disease.*

Female Genital Rhabdomyosarcoma

- **Endoscopic biopsy plus chemotherapy controls the large majority**. Lymph node biopsy is not recommended; **PRE/DPE only if R0 resection is anticipated**; **hysterectomy or exenteration only for chemotherapy-unresponsive tumours**.
- **Initial approach for all three sites**: abdominopelvic MRI → bimanual examination under anaesthesia (EUA) → cystoscopy → vaginoscopy with endoscopic biopsy.

VULVOVAGINAL, CERVICAL AND UTERINE RHABDOMYOSARCOMA COMPARED

	Vulvovaginal	Cervical	Uterine
Share of female GU-RMS	Commonest (about two-thirds)	About one-fifth	About one-tenth
Features	Fusion-negative ERMS commonest; prolapsing mass	Small pedunculated tumours	Younger; worse OS; more stage IV despite favourable site
Biopsy	Endoscopic	Transvaginal polypectomy or excisional biopsy; partial cervical excision by loop electrosurgical excision procedure (LEEP) of choice when feasible	Endoscopic; percutaneous/transabdominal needle biopsy for large tumours
Nodes	Pelvic lymph node dissection not recommended — nodal spread rare	Lymph node biopsy not recommended	—
Reassessment	MRI and vaginoscopy after 3 and/or 6 cycles; incomplete response → repeat biopsy; positive → radiotherapy (ideally brachytherapy)	Endoscopic biopsies ($\pm$ hysteroscopy) after induction even with no visible disease — unlike vulvovaginal or uterine	Complete response → no benefit from DPE

	Vulvovaginal	Cervical	Uterine
Residual disease	Surgery only if R0 achievable	Negative biopsies → surveillance; residual → brachytherapy or trachelectomy (excellent local control, surprisingly good fertility)	Open or laparoscopic hysterectomy, R0 only ; preserve vagina and ovaries (oophorectomy only for gross ovarian involvement); no vaginal hysterectomy (poor visualisation); external beam radiotherapy (EBRT) for positive margins
Exenteration	Discouraged	—	Generally discouraged

KEY POINTS

- Cystoscopic biopsy and catheter drainage are the best initial choices in BP-RMS.
- Most BP and female GU-RMS responds to chemotherapy and radiation without extirpative surgery.
- All PT-RMS needs inguinal orchiectomy; PT-RMS with N1 disease or age over 10 needs RPLND.

Concept — female GU-RMS is **cured by chemotherapy with organ preservation**; surgery is reserved for residual disease that can be cleared to R0, and the cervix is the exception demanding biopsy even after an apparent complete response.

Perineal and Perianal Rhabdomyosarcoma

- **Very rare; alveolar histology more common**, about 80% fusion-positive. **Poor outcomes** (5-year OS about one-half); larger tumours and older patients fare worse. A significant minority of survivors have **faecal incontinence**; some urinary frequency or incontinence.

Special Situations (e.g., Infants)

- Infants have **greater treatment-related morbidity** and a higher probability of a predisposition syndrome.
- **Local control is the problem**: the infant pelvis limits surgery, and EBRT is often withheld for fear of damaging the bony pelvis and hips → **higher local failure rates**, likely from reduced radiotherapy.
- COG and European protocols **allow deviation from protocol radiotherapy for fusion-negative children under 24 months**; therapy for infants, especially congenital or neonatal RMS, remains controversial.

Late Effects of Rhabdomyosarcoma Treatments

- Survivorship is a key focus of current protocols; the **COG Late Effects Committee** publishes a comprehensive guide.
- Complications are frequent and often severe: **second malignancy, renal and urinary dysfunction, hormonal disturbance, infertility and sexual dysfunction, bowel complications, psychological disturbance**.

ACUTE TOXICITIES OF RHABDOMYOSARCOMA CHEMOTHERAPY

Agent	Signature acute toxicity
Vincristine	Neuropathy
Doxorubicin	Heart failure
Cyclophosphamide	Myelosuppression; haemorrhagic cystitis

- **Radiotherapy** is a major causative factor — almost **tenfold more adverse sequelae** than in untreated patients.
- **Sexual and reproductive**: in a long-span survivor series urinary complaints were uncommon but **sexual limitation was very common in both sexes**. Female pelvic RMS survivors followed to 20 years had near-

universal delayed morbidity — **endocrine/ovarian dysfunction** and **gynaecological dysfunction** (fistulae, vaginal stenosis) — and births were rare (the only one in a non-irradiated patient).

- **Bladder:** impaired by surgery, chemotherapy or radiation; **more than 60% of BP-RMS patients have some dysfunction despite bladder preservation**. **Detrusor–sphincter dyssynergia** was commonest in one small urodynamic series; detrusor overactivity and reduced capacity in others.
- **Second malignant neoplasm:** radiotherapy and chemotherapy (particularly **anthracyclines** and **alkylating agents**) raise risk; pelvic RMS survivors develop second malignancy at **about six times population rates** → **long-term follow-up in specialised survivorship clinics**.
- **Infants** carry the heaviest burden: in the CWS experience most developed late toxicity, **commonest renal**.

KEY POINTS

- Survivorship concerns are significant: secondary malignancy, bladder function and sexual health.

*Concept — the survival gains were bought with **radiation and alkylator-heavy chemotherapy to organs that are still growing**; bladder preservation is anatomical, not functional, and that is why follow-up is lifelong.*

OTHER PAEDIATRIC BLADDER TUMOURS

Urothelial Carcinoma/Papillary Urothelial Neoplasms of Low Malignant Potential

Presentation, Diagnosis and Management

- Most paediatric urothelial tumours are **papillary urothelial neoplasms of low malignant potential (PUNLMP)** or **low-grade papillary urothelial carcinoma**, depending on the classification used.
- Most present with **haematuria** and are diagnosed on renal–bladder ultrasound (RB-US) or CT, typically as a papillary tumour of the bladder wall.

Treatment

- **Almost exclusively endoscopic:** transurethral resection of bladder tumour (TURBT) with a 9- or 11-French paediatric resectoscope, or **cold-cup biopsy if the stalk is narrow**.
- As in adults, **fulgurate the base** ($\pm$ deep base biopsy). The authors instil immediate post-resection intravesical chemotherapy by institutional norm; there are no data for or against.

Outcomes

- Adults: PUNLMP **recurs in about a third** after complete resection; some enlarge if unresected; **malignant transformation or progression to high grade is exceedingly rare**.
- Children: similarly **low transformation, moderate recurrence risk** → follow-up needed. In the authors' experience recurrence after complete resection is rare (their single recurrence was in an adolescent with **Lynch syndrome**) → **surveillance by RB-US**.

Special Situations

Spina Bifida/Bladder Augmentation

- Original hypothesis: **mixing urine and faeces is carcinogenic**; enterocystoplasty, by separating them, should avoid cancer. Aggressive cancers were then reported in reconstructed bladders, especially after **gastrocystoplasty (survival about 50%)** → proposals for **routine endoscopic screening and a moratorium on gastrocystoplasty**.

- These recommendations were then challenged:
 - **Markov decision analysis:** even under aggressive assumptions, **annual cytology and/or cystoscopy is neither effective nor cost-effective** — spina bifida-related cancers progress within months of a normal surveillance study.
 - **Matched comparison** of intestinal (ileal/colonic) augmentation versus non-augmented controls: **no significant increase** in cancer risk or characteristics.
 - **Systematic review:** augmented and non-augmented patients have **similar, poor survival**.
- Conclusion: (1) **neurogenic bladder carries a baseline lifetime cancer risk of 2–4% whether or not augmented;** (2) **these cancers are aggressive;** (3) **screening asymptomatic patients is neither effective nor cost-effective.**

Bladder Exstrophy

- As with spina bifida, **exstrophy carries increased bladder cancer risk:** adenocarcinoma, squamous cell and mixed tumours are reported.
- These cancers are **rare:** few respondents in a national exstrophy survey reported cancer (likely undercounted because deaths remove respondents, but reassuring).
- Cancer also arises in **non-augmented** patients; lifetime risk in augmented patients about 2%; **no compelling evidence that augmentation changes the likelihood or severity** of bladder cancer.

KEY POINTS

- Paediatric urothelial tumours are rare and typically low-grade/PUNLMP.
- Recurrence is rare and should prompt suspicion of a predisposition syndrome.
- Spina bifida and exstrophy carry a 2–4% baseline bladder cancer risk not increased by augmentation.

Concept — the cancer risk belongs to the abnormal bladder, not to the augment; and because these tumours are aggressive and fast-growing, periodic screening cannot catch them in time.

Nonmalignant Bladder Lesions

NONMALIGNANT PAEDIATRIC BLADDER LESIONS COMPARED

Lesion	Nature and associations	Presentation and diagnosis	Treatment and course
Inflammatory myofibroblastic tumour (IMT)	Rare mesenchymal neoplasm: spindle-cell proliferation within an inflammatory infiltrate ; median age 9.5 years; EpSSG class it as intermediate malignancy (non-RMS soft-tissue sarcoma family) because it can spread in older patients; most express anaplastic lymphoma kinase (ALK) rearrangement	Bladder-wall mass on US and CT	Complete surgical resection is standard for localised disease (bladder IMT almost always resectable, behaves benignly); no standard for advanced disease; excellent EFS and OS regardless of ALK status
Nephrogenic adenoma	Benign papillary lesion from bladder trauma : recurrent infection, radiation, bladder surgery including renal transplantation	Haematuria, dysuria, frequency, recurrent urinary tract infection (UTI), detrusor overactivity; cystoscopic biopsy required	Transurethral resection or fulguration plus treat the cause (antibiotic prophylaxis, improved voiding); recurrence common → long-term follow-up ; reported malignant transformation is rare and of doubtful initial diagnosis

Lesion	Nature and associations	Presentation and diagnosis	Treatment and course
Eosinophilic cystitis	Rare in children, as in adults	Haematuria, overactive bladder, dysuria, flank/abdominal pain, occasionally obstruction; cystoscopic biopsy essential — no typical imaging or endoscopic appearance; eosinophilia and eosinophiluria usually absent	Medical therapy (steroids, antihistamines, antibiotics) in most; TURBT or partial cystectomy occasionally
Haemangioma	Sporadic or syndromic (Sturge-Weber, Klippel-Trenaunay-Weber); about 30% have haemangiomas elsewhere; cavernous commonest , then capillary, arteriovenous	Gross, sometimes severe, haematuria ; may mimic malignancy	Propranolol or endoscopic laser ablation can cure; partial cystectomy for large tumours or endoscopic failure
Fibroepithelial polyp	Rare benign lesion at any age, anywhere in the urinary tract	Obstruction once large; pelvis/proximal ureter → mimics pelviureteric junction obstruction with Dietl crisis; urethra → mimics urethral valves , especially in infants	—

- What the table tests: **biopsy before definitive treatment** (nephrogenic adenoma and eosinophilic cystitis cannot be diagnosed on imaging), and the **mimics** — haemangioma for malignancy, fibroepithelial polyp for pelviureteric junction obstruction or valves.

KEY POINTS

- Most nonmalignant bladder tumours are managed medically, endoscopically or by partial cystectomy.

TESTICULAR TUMOURS

- **1–2% of paediatric solid tumours**; incidence about **1 per 100,000 under 15 years**. Most occur at **2–4 years**, with a **second rise at puberty**.
- **Benign tumours are far commoner than in adults** (38–74% of paediatric testicular tumours). **Germ cell tumours (GCTs) comprise the vast majority** at all ages; the rest are stromal.
- **Teratoma is the commonest prepubertal benign tumour; yolk sac tumour (YST) the commonest prepubertal malignant tumour.**

Paediatric Germ Cell Tumours

PREPUBERTAL TESTICULAR GERM CELL TUMOURS COMPARED

	Teratoma	Epidermoid cyst	Yolk sac tumour
Frequency	Commonest prepubertal tumour	Uncommon	Less common than teratoma; commonest malignancy of infants and toddlers
Behaviour	Benign before puberty	Monophasic teratoma (ectoderm only); benign prepubertal; rare germ cell neoplasia in situ (GCNIS) nearby postpubertal	Malignant; 90% of prepubertal YST are stage I

	Teratoma	Epidermoid cyst	Yolk sac tumour
US	Anechoic or complex cystic lesions with highly echogenic foci	"Onion-skin" concentric keratin rings	—
Markers	Normal	Normal	AFP almost always raised
Surgery	Partial orchiectomy if markers normal and frozen section benign ; otherwise radical	Partial orchiectomy , particularly if bilateral; frozen section if postpubertal	Radical inguinal orchiectomy — usually curative
After surgery	Prepubertal: no further oncological evaluation or follow-up	—	Follow-up as adult National Comprehensive Cancer Network (NCCN) guidance for nonseminomatous GCT (NSGCT)

• **YST staging and escalation:**

- **CT chest, abdomen and pelvis** for complete staging. After orchiectomy, a raised AFP should fall appropriately if there is no metastasis — remembering **physiological AFP elevation in infants**.
- **Stage I prepubertal: no RPLND and no chemotherapy.**
- **Prior scrotal biopsy → stage II**, but delayed scrotal resection is generally unnecessary because these children receive systemic chemotherapy.
- **Lymphadenopathy with normal markers → node sampling or biopsy** to confirm metastasis. **Lymphadenopathy with raised markers = stage III**; stage III and IV receive **cisplatin, etoposide and bleomycin**, as in adults.
- Prepubertal persistent disease after chemotherapy → **biopsy or RPLND** to confirm histology and guide adjuvant chemotherapy.

Presentation and Diagnosis

- Typically a **painless testicular mass**; rarely found with epididymitis, torsion, hydrocele, undescended testis or hernia.
- As in adults: examination, history for risk factors, scrotal US and markers — **lactate dehydrogenase (LDH), AFP, beta-human chorionic gonadotrophin (β-hCG)**. **Unlike adults, check hormone levels if a hormonally active tumour is suspected** (precocious puberty).

Imaging.

- **Doppler US**; it **cannot reliably distinguish benign from malignant**. Anechoic cystic lesions suggest benignity (simple cyst, cystic dysplasia, teratoma). Epidermoid cyst: heterogeneous mass with alternating hypoechoic and hyperechoic keratin rings (onion skin). Benign masses typically show **absent flow in surrounding tissue** from compression.
- Malignancy suspected → **CT chest, abdomen and pelvis**; timing before or after resection is controversial (authors: before).

Biomarkers.

ALPHA-FETOPROTEIN VERSUS BETA-HUMAN CHORIONIC GONADOTROPHIN

	AFP	β-hCG
Nature and source	Single-chain glycoprotein of fetal yolk sac, liver and gastrointestinal tract	Glycoprotein from embryonal carcinoma and mixed teratomas
Half-life	5 days	24 hours
Prepubertal tumours	YST invariably produces AFP ; teratoma and benign lesions normal	Rarely raised

	AFP	β-hCG
Caveat	Physiologically raised in infancy; reaches the adult range (<10 ng/mL) by 6–9 months	—

- **Benign-appearing US lesion + negative AFP/β-hCG → attempt partial orchidectomy.**
- After removal, staging uses pathology, extent of local and metastatic disease and marker levels; marker-positive tumours are followed for an **appropriate half-life decline** after radical inguinal orchidectomy.

Special Situations

Leukaemia and lymphoma.

- The **commonest childhood malignancies to spread to the testis**. With modern intensive chemotherapy, testicular relapse in acute lymphoblastic leukaemia (ALL) is **2% or less** → biopsy, orchidectomy or testicular irradiation is exceedingly rarely indicated.

Disorders of sex development.

- Data are scant despite research dating to the 19th century. The **pooled risk of 12% is misleading** — risk varies widely by diagnosis.
- **Y-chromosome material**, particularly the sex-determining region Y (SRY) and gonadoblastoma on Y chromosome (GBY) regions, **significantly raises risk**; XY disorders of sex development (DSD), especially complete and partial gonadal dysgenesis, carry very high risk.

TABLE 64.4 MALIGNANCY RISK OF DSD DIAGNOSES

Risk group	DSD diagnosis	Table 64.4 (%)	Body text (%)
High	Complete gonadal dysgenesis	35	33
	Partial gonadal dysgenesis	24	24
	Partial androgen insensitivity	15–50	30–50
	Frasier syndrome	60	30–50
	Denys–Drash syndrome	50	30–50
	Turner syndrome with Y	30	30–50
Intermediate	Mixed gonadal dysgenesis	8	—
	Complete androgen insensitivity	1–6 (gonadoblastoma, Sertoli cell)	—
	Ovotesticular DSD	1–25	—
	Klinefelter syndrome	14 (Leydig cell)	—
Low	Turner without Y; 5-alpha-reductase deficiency; 46,XX DSD	Case reports only; about baseline population risk	

Data compiled from EAU guidelines, 2023; Kosti et al., 2019; Pyle et al., 2017; Slowikowska-Hilczer et al., 2020.

- The two columns differ because **the table pools several sources including European Association of Urology (EAU) guidance, whereas the text cites a single cohort (gonadal dysgenesis) and a combined 30–50% range** for the other high-risk diagnoses.
- Pattern: **dysgenetic gonads with Y material are high risk; well-formed gonads (androgen insensitivity complete, Klinefelter) are intermediate; no Y material is low**. Trap: complete androgen insensitivity is **not** high risk; partial is.
- Tumour types: **gonadoblastoma** — potentially premalignant, analogous to GCNIS, **frequently bilateral**; large seminoma-like germ cells with stroma resembling immature granulosa/Sertoli cells. When invasive it

becomes **dysgerminoma**, histologically similar to seminoma (e.g. arising from streak gonads in a 46,XY phenotypic female).

- **Gonadectomy is controversial**, especially for undescended gonads; it weighs tumour risk, gender identity and fertility wishes (fertility potential is itself sometimes debatable) → **personalised decision after lengthy, informed discussion** of risk tolerance.

Testicular microlithiasis.

- Controversial across all ages. An EAU Paediatric Urology Guidelines panel systematic review of several hundred children with testicular microlithiasis (TM) found **only one testicular cancer through pubertal follow-up**, even though many had risk factors (sub/infertility, cryptorchidism, previous testicular cancer).
- **Recommendation: monthly self-examination from puberty only for TM with significant risk factors; no further childhood surveillance.** Whether workup is needed at transition to adulthood is unclear.

KEY POINTS

- Scrotal masses in children present, and are evaluated, as in adults.
- Unlike adults, partial orchidectomy is the most appropriate surgical option for most prepubertal testicular tumours, unless markers are raised or frozen section shows a GCT.
- DSD raises testicular tumour risk; some diagnoses carry higher risk than others.

Stromal Tumours

Specific Tumour Types

TESTICULAR STROMAL TUMOURS IN CHILDREN COMPARED

	Leydig cell	Sertoli cell	Juvenile granulosa cell	Testicular adrenal rest (TART)
Frequency	Commonest stromal tumour (children and adults)	Second commonest in children; commoner in adults	Rare in testis; commonest ovarian stromal tumour	Peripubertal boys with congenital adrenal hyperplasia (CAH)
Presentation	Mass, often precocious puberty (may not resolve after surgery)	Mass, scrotal pain, gynaecomastia ; markers rarely abnormal	Solid mass	Palpable mass or scrotal pain; undertreated CAH (high adrenocorticotrophic hormone, ACTH) , often after growth spurts
Pathology	Well encapsulated, yellow-brown (steroid); resembles adrenal rest; Reinke crystals pathognomonic, present in about 40%	Solid, white-yellow, lobulated, no haemorrhage or necrosis; large polygonal eosinophilic cells (large ones mimic Leydig)	Gonadal stromal origin	Differs from Leydig by low mitotic activity, extensive fibrosis; synaptophysin immunohistochemistry particularly helpful
Prepubertal behaviour	Benign ; no malignant features reported	Metastasis uncommon; recurrence after partial orchidectomy exceptional; contralateral recurrence about 1%	No malignant cases in a large review	Can cause primary gonadal dysfunction

	Leydig cell	Sertoli cell	Juvenile granulosa cell	Testicular adrenal rest (TART)
Treatment	Partial orchidectomy	Partial orchidectomy; no adjuvant/ neoadjuvant regimen or RPLND benefit	Partial prepubertal; postpubertal more aggressive (radical orchidectomy ± chemotherapy), as some adult tumours metastasise	Increase steroid dose; some need partial orchidectomy to restore endocrine function

Treatment

Surgery.

- **Normal preoperative markers → testis-sparing partial orchidectomy**, as for GCTs. Multiple systematic reviews give **no justification for radical orchidectomy in prepubertal boys**.
- Because **scrotal US cannot reliably separate stromal from germ cell tumours**, the urologist should proceed confidently with partial orchidectomy in a prepubertal child with normal markers.
- Leydig, Sertoli and juvenile granulosa cell tumours behave more aggressively after puberty → **radical orchidectomy is likely justified in Tanner 5 adolescents and young adults**.

Systemic treatment.

- **Chemotherapy is rarely indicated:** these tumours are rarely malignant before puberty, surgery alone usually suffices, and there are **little to no data for any regimen**.

Outcomes

- **Generally excellent:** recurrence 1–7%, metastasis rarely reported in prepubertal children, versus **about 10% aggressive behaviour in adults**.

KEY POINTS

- Partial orchidectomy is the most appropriate surgical option for testicular stromal tumours in prepubertal children.

Concept — puberty is the dividing line: prepubertal testicular tumours are overwhelmingly benign, so normal markers license testis-sparing surgery for germ cell and stromal lesions alike; after puberty, adult biology and adult radicalism return.

Sacroccygeal Teratoma

- Sacroccygeal teratoma (SCT): rare pelvic GCT arising from the coccyx, incidence about **1 in 40,000**; **female predominance**; many diagnosed prenatally.
- Often managed by paediatric surgeons, but **pelvic by definition and treatment affects the bladder → paediatric urologists must be involved**.
- **Neurogenic bladder in 36–51%**, commonly presenting as **urinary retention**; most need intervention, from clean intermittent catheterisation (CIC) to bladder reconstruction.
- **Altman classification:** type I (predominantly external, minimal sacral involvement) is commonest; **types II–IV (significant internal abdominopelvic component) carry much higher neurogenic bladder and bowel risk**.
- Proposed risk-based monitoring: **annual RB-US for all until toilet training; UDS for higher Altman types, neurogenic signs or RB-US abnormality**.

- **Currarino syndrome** triad: **anterior sacral mass** (usually meningocele; can be SCT or dermoid) + **bony sacral defect** + **anorectal malformation** → **universal neurogenic bladder**, lifelong surveillance and active management.

KEY POINTS

- SCT carries high neurogenic bladder risk, particularly in complex (internal) tumours and Currarino syndrome.

Paratesticular Tumours

- **RMS is by far the commonest paratesticular tumour.** Others: epididymal cysts (common), adenomatoid tumour, **epididymal cystadenoma (classically von Hippel–Lindau, VHL, syndrome; also sporadic)**, Müllerian cysts.
- **Testis-sparing excision** is generally appropriate, but **an inguinal approach is mandatory and frozen section must be confirmed before closure**, because PT-RMS may be misdiagnosed.

OTHER FEMALE GENITOURINARY TRACT TUMOURS

- With more fellowship-trained paediatric gynaecologists, urologists are not always primarily responsible, but ovarian and adnexal lesions surface as **incidental findings or during symptom workup**, so familiarity is required.

Cystic Lesions

- **Up to 98% of female newborns have small ovarian cysts on US;** clinically significant newborn cysts are rare (about 1 in 2500 births); 2–3% of preschool and school-age girls have incidental cysts.
- Cysts reflect **maternal hormones and involute spontaneously** as levels fall. Intervene only if **indeterminate or symptomatic**.
- **Prepubertal surgical indications:** persistence, concern for malignancy, or any symptom including hormonally induced changes.
- **Adolescents:** cysts usually from dysfunctional ovulation; aim is **ovarian preservation**; hormonal manipulation common; surgery for non-resolution, persistent symptoms, or concern for teratoma or malignancy. Simple cysts causing mass effect or discomfort → **organ-sparing surgery**.
- Malignant cysts are rare; **even complex cysts are often observed**.

Ovarian Malignancies

- About **1% of childhood cancers**; incidence 2.6 per 100,000 females. Classified, as in the testis, into **germ cell, stromal and epithelial**; **epithelial tumours are very rare before puberty**.
- Imaging: **abdominal US with a full bladder** → **abdominopelvic MRI**.

PAEDIATRIC OVARIAN TUMOURS BY ORIGIN

	Germ cell	Stromal (sex cord)	Epithelial
Types	Seminoma equivalent: dysgerminoma (adolescents; undifferentiated germ cells). Nonseminoma: YST (endodermal sinus), teratoma (mature, immature), embryonal carcinoma, choriocarcinoma; gonadoblastoma in dysgenetic gonads (rarely metastasises)	Theco-fibroma, Sertoli, Leydig, juvenile granulosa cell tumour (JGCT) , Sertoli–Leydig cell tumour	Cystadenoma (benign or malignant); small cell carcinoma (extremely rare)

	Germ cell	Stromal (sex cord)	Epithelial
Age / presentation	Bimodal: infancy and late adolescence ; pain, distension, mass, acute abdomen, torsion, precocious puberty	Hormonally active: JGCT → oestrogen → precocious puberty; Sertoli-Leydig → androgens → virilisation , pain, mass	Mostly older patients
Key facts	Teratoma commonest ; teratomas contain all three germ layers; mature teratoma (dermoid) benign → resect; immature teratoma and YST aggressive → COG staging and treatment ; markers AFP, β-hCG, LDH (YST, embryonal, choriocarcinoma)	Exceedingly rare malignancies; screen for DICER1 syndrome ; low stage → surgery alone; higher stage → chemotherapy/multimodal	—
Prognosis	Very good ; OS above 90% for stages I and II	Generally favourable	—

Nonrhabdomyosarcoma Vulvovaginal Tumours

- **Primary vulval YST** is reported in female toddlers (usually an adult tumour); these may instead be **SMARCB1-deficient vulvar neoplasms**, a group including epithelioid sarcoma. Very rare, **poor prognosis** → excision or biopsy with **regional node dissection**, then **aggressive chemotherapy and radiotherapy**.

Urethral Prolapse

- **Frequently mistaken for RMS**: a **doughnut-like circumferential swelling around the urethral meatus**; in the United States most often in **prepubertal Black girls**.
- **Topical steroid or oestrogen cream is frequently curative**; surgical excision rarely.
- **Neither cancerous nor precancerous** → once diagnosed, no further workup or monitoring.

KEY POINTS

- Ovarian cysts are common; ovarian cancers in girls are rare.
- Girls with Sertoli-Leydig cell tumours should be screened for DICER1 mutations.

Concept — the female gonad follows the **same germ cell–stromal–epithelial logic as the testis**; most lesions are hormonal or benign, so **ovarian preservation** is the default and the stromal tumours point back to DICER1.